DATAQUEST — STAGE 3

Data Detective: Analyze, Visualize & Present

KG Reddy College of Engineering & Technology | Data Science Club | IKARUS 2026

PART A — PARTICIPANT BRIEF

Challenge format: Team-based data analysis activity. Teams receive a small dataset, inspect and clean it, calculate useful metrics, create a clear visualization, and present evidence-based findings.

Recommended duration	**60 minutes**
Total marks	100
Recommended team size	2–3 participants
Permitted tools	Python (Pandas/Matplotlib), Excel, Google Sheets, or equivalent
Deliverables	Cleaned data/table, 2–3 key metrics, one chart, and a 3-minute presentation

Scenario

A college canteen manager wants to understand daily sales performance and customer preferences. Your team has been given the sample sales records below. Analyze the data and recommend two practical actions that could improve canteen operations. Use only evidence that can be supported by the dataset.

Dataset: canteen_sales.csv

Date	Category	Item	Units_Sold	Unit_Price	Rating
2026-09-01	Snacks	Samosa	30	15	4.2
2026-09-01	Drinks	Lemon Juice	18	25	4.0
2026-09-02	Meals	Veg Meal	22	70	4.5
2026-09-02	Snacks	Sandwich	16	40	4.1
2026-09-03	Drinks	Lemon Juice	25	25	4.3
2026-09-03	Snacks	Samosa	35	15	4.0
2026-09-04	Meals	Veg Meal	20	70	4.4
2026-09-04	Drinks	Tea	28	10	3.8
2026-09-05	Snacks	Sandwich	21	40	4.4
2026-09-05	Drinks	Tea	32	10	3.9
2026-09-06	Meals	Veg Meal	26	70	4.6
2026-09-06	Snacks	Samosa	40	15	4.3

Tasks

Task 1 — Data inspection and preparation (15 marks)

- Check whether any values are missing, duplicated, or invalid.
- Add a Revenue column using Units_Sold × Unit_Price.
- State any cleaning decisions you made.

Task 2 — Key metrics (25 marks)

- Calculate total revenue for all records.
- Find the category with the highest total revenue.
- Find the item with the highest total units sold.
- Calculate the average customer rating.

Task 3 — Visualization (20 marks)

- Create one chart that helps the manager compare revenue by category or item.
- Include a meaningful title, labelled axes, and readable category names.
- Write one sentence explaining what the chart shows.

Task 4 — Insights and recommendations (20 marks)

- Write two insights supported by specific numbers from the data.
- Recommend two actions for the canteen manager.
- Mention one limitation of the dataset; do not claim that the sample proves a cause-and-effect relationship.

Task 5 — Team presentation (20 marks)

- Present for up to 3 minutes, followed by up to 2 minutes of questions.
- Every recommendation must be linked to a metric, chart, or observed pattern.
- All team members should be ready to explain the analysis.

Submission checklist

- Cleaned dataset or summary table
- Revenue calculation and requested metrics
- One chart
- Two evidence-based insights and two recommendations
- Short presentation (slides or one-page summary)

PART B — CONFIDENTIAL ORGANIZER ANSWER GUIDE

Keep this section separate from the participant brief during the event. Values below are computed from the sample dataset.

Expected calculations

Total revenue: ₹9,490.

Revenue by category:

- Meals: ₹4,760
- Snacks: ₹3,055
- Drinks: ₹1,675

Units sold by item:

- Samosa: 105 units
- Veg Meal: 68 units
- Tea: 60 units
- Lemon Juice: 43 units
- Sandwich: 37 units

Average rating across records: 4.21 / 5.

Total units across all records: 313.

Example acceptable insights

- Meals contribute the highest category revenue (₹4,760) in this sample, largely because the unit price is higher.
- Samosa has the highest total units sold (105 units), so the team may recommend monitoring stock levels.
- Tea has comparatively low ratings in the sample (3.8 and 3.9); the team could recommend collecting more feedback before changing the recipe.
- The dataset covers only six dates and a limited set of products, so it may not represent a full month or typical demand.

Marking rubric (100 marks)

Criterion	Marks	What judges look for
Data inspection, cleaning, and revenue column	15	Correct checks; reasonable handling; accurate revenue values
Metrics and calculations	25	Correct totals, category/item aggregation, average rating
Chart quality and interpretation	20	Correct chart, labels/title, useful comparison, accurate interpretation
Insights and recommendations	20	Evidence-backed insights; practical recommendations; limitation acknowledged
Presentation and responses	20	Clear structure, teamwork, timing, ability to explain methods

Tie-break and fairness guidance

- First tie-break: correctness of metrics and revenue calculations.
- Second tie-break: strength of evidence and clarity of chart.
- Third tie-break: practical recommendations and response to judge questions.
- Accept different tools and chart types when calculations and explanations are correct.
- Do not award full insight marks for unsupported claims or conclusions that imply causation.

Organizer checklist

- Provide the dataset to every team in the same format.
- Announce allowed tools, time limit, deliverables, and presentation time before starting.
- Keep the answer guide confidential.
- Use the same rubric for all teams and record marks per criterion.
- If using a larger hidden dataset, validate its values and answer key before the event.